

R-LearnXR: Reproducible R-Based Virtual Laboratories for Data Science Education

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Working submission draft. This document follows the R Consortium ISC proposal template revised 2026-03-24. Named community validation and final signatories must be added before submission.

Executive Summary

R-LearnXR will provide open infrastructure for creating reproducible, browser-based virtual data laboratories with R and Quarto. R educators can readily publish code, text, and two-dimensional graphics, but an interactive multivariate lesson still requires authors to coordinate analysis data, browser rendering, learner prompts, accessible alternatives, environment locking, and publishing. Existing examples are often tied to one course or an opaque hosted service and are difficult for another instructor to adapt.

The project will turn a working prototype into a community-ready R package and lesson framework. Learners predict a result, edit and execute real R code locally through a pinned WebR runtime, inspect the returned data in a keyboard-operable 3D scene or semantic table, explain coordinate evidence, complete a transfer task, and export reproducible R or Quarto source. Authors will scaffold lessons, render scenes from R data frames, and run automated checks for deterministic seeds, environment locks, data licenses, accessible alternatives, and required learning-loop elements.

Grant work will stabilize the package API, add a third openly licensed example lesson, expand reproducibility and accessibility checks, conduct a small educator and learner pilot, and publish contributor training and maintenance documentation. The prototype reduces delivery risk by demonstrating the complete browser workflow now. We request **\$10,000 for six months, January through June 2027**. Existing prototype development is in-kind evidence of feasibility and is not included as retroactive cost.

Signatories

Project team

- **Jewoong Moon, Project Lead and primary contractor.** Responsible for learning-experience design, R and Quarto integration, reproducibility workflow, documentation, community coordination, and project reporting. Track record and current implementation are available in the [public R-LearnXR repository](#).
- **Co-lead or core project team member: to be confirmed before submission.** The selected person will bring R package maintenance, statistics education, accessibility, or empirical learning-evaluation experience; independently review delivery; oversee the pilot; and own at least one milestone outcome.

Contributors

People who contribute lesson examples, accessibility review, code, or proposal feedback will be named here after confirming how they wish to be credited.

Consulted

At least two R or Quarto educators or maintainers will review the need, scope, API, and reuse plan before submission. Attributable feedback or public evidence will be linked here. No external endorsement is claimed in this working draft.

The Problem

R instructors who want learners to reason about multivariate results must assemble several systems: R data preparation, interactive rendering, explanatory prompts, accessible alternatives, reproducible environments, and publishing. These pieces are usually built ad hoc. The result is duplicated effort, fragile course examples, and activities that other instructors cannot readily inspect or reuse.

The problem affects instructors, package authors, workshop organizers, contributors, and learners. Instructors need a maintainable route from an R analysis to an interactive lesson. Learners need guided evidence-based reasoning instead of a visualization that is merely viewed, plus a non-pointer path when a 3D canvas is inaccessible. Maintainers need examples that can be checked automatically when data, packages, browsers, or publishing tools change.

R already offers strong analysis, visualization, Shiny, WebGL, WebAssembly, and publishing capabilities. R-LearnXR does not replace these projects. It supplies a focused educational layer connecting R-generated data, an explicit learning workflow, accessible evidence inspection, reproducibility checks, and contribution practices. The browser-first design makes headsets optional and avoids proprietary application servers.

Solving this problem will let the R community share virtual laboratories as open educational resources. An instructor will be able to scaffold a documented template, map three variables from an R data frame into an inspectable scene, add prediction and explanation prompts, verify the build, and publish through an ordinary Quarto workflow. Contributors can improve the package, adapt examples to new domains, translate lessons, and reuse the checks in other educational projects.

The Proposal

Overview

R-LearnXR will become a small R package and Quarto lesson framework for turning analysis results into inspectable learning evidence. R remains the source of data and analysis. A pinned WebR runtime lets learners execute edited R code locally in the browser. The result updates a portable 3D scene, semantic table, observation text, artifact hash, and reproducibility record. Quarto provides the surrounding lesson and publishing workflow.

The grant will fund the future work needed to move from prototype to community infrastructure: API stabilization, reusable authoring primitives, a third authentic example, stronger checks, external validation, contributor onboarding, and a documented release process.

Minimum viable product

1. A tested R package with stable functions to scaffold lessons, render browser scenes, and check lesson projects.
2. A Quarto template implementing Orient, Predict, Run R, Explore, Explain, Transfer, and Reproduce.
3. Three openly licensed example lessons using public datasets.
4. Automated checks for data licenses, deterministic seeds, environment locks, portable paths, required learning-loop elements, accessible alternatives, and generated-artifact hashes.
5. An authoring guide, accessibility guide, contributor pathway, pilot protocol, and issue templates.
6. Continuous integration for package tests, lesson checks, and Quarto rendering.

Architecture and dependencies

R prepares and validates author-supplied data. The package writes a small JSON artifact and a maintained HTML/JavaScript laboratory. WebR executes learner-edited R, validates a returned **scene** data frame, and synchronizes it with the canvas and semantic table. Quarto embeds or links the laboratory. A checker produces human-readable reports, and continuous integration repeats the checks and renders examples.

Core dependencies are R, Quarto, WebR, standard browser APIs, **testthat**, and openly licensed datasets. The project will pin tested versions and document every external asset. No account, proprietary service, headset, or application server is required for learners.

Assumptions, risks, and recovery

- **3D is useful only for selected multivariate tasks.** The pilot will test task fit. The framework will retain table-first and two-dimensional alternatives when 3D adds no instructional value.
- **A pinned WebR runtime remains maintainable.** The release will expose loading and error states and evaluate self-hosted runtime assets. Browser execution is an enhancement; author-side rendering and accessible artifacts remain available.
- **Educators can author concise evidence prompts.** Contributor training will include editable prompt patterns and a review rubric.
- **Scope could expand into a general XR platform.** The grant excludes headset-native applications, multiplayer systems, accounts, cloud analytics, and a general-purpose 3D grammar.
- **Early adoption may be limited.** Success will use documented reuse intent, pilot completion, outside contribution, and maintainable artifacts, not inflated usage counts.

Project Plan

Start-up and technical delivery

The public repository, licenses, contribution guide, code of conduct, prototype lessons, and continuous-integration workflow already exist. During the first month, the team will confirm roles, publish reporting checkpoints, convert remaining prototype decisions into a release schema, and open milestone issues for community review.

Budgeted deliverable package	Target	Grant-period outcome	Budget
1. Package and API hardening	Jan. 31, 2027	Stable API, schema, tests, and release criteria	\$2,200
2. Reusable lesson system	Mar. 15, 2027	Quarto authoring primitives and a third authentic lesson	\$2,300
3. Reproducibility and accessibility	Apr. 15, 2027	Machine-readable checks, CI evidence, and keyboard/table validation	\$2,000
4. External validation and onboarding	May 31, 2027	Educator and novice-learner pilots, fixes, and contributor training	\$2,000
5. Community release and reporting	Jun. 30, 2027	Tagged release, documentation, final report, and maintenance roadmap	\$1,500
Total			\$10,000

The rows identify scope, outcomes, and partial-funding priorities. The budget assumes the R Consortium’s standard disbursement of 50% at contract signing and 50% after accepted completion; it does not request five separate payments. Funds support direct labor during the grant period. No indirect costs, travel, lodging, food, publication fees, or personal hardware are requested.

Community availability and dissemination

Code will remain MIT licensed. Original educational content and templates will use CC BY 4.0, and datasets will retain explicit source licenses. Work will be developed in public issues with contribution templates and quarterly progress notes. The team will solicit feedback from R educators, maintainers, and an R user group or comparable community, prepare an R Consortium blog update, and publish an end-of-project report that connects pilot findings to resolved issues and remaining work.

Success

Definition of done

- Package functions and lesson checks pass in continuous integration.
- Three complete lessons render from documented, locked environments.
- Every example supports keyboard operation and a semantic data alternative.
- Authoring, accessibility, contribution, validation, and maintenance documentation is public.
- A tagged release and final project report are archived and linked from the repository.

Measuring success

- At least two instructors or maintainers document a concrete review or reuse outcome.
- At least one structured pilot is completed and summarized without exposing private learner data.
- At least 80% of novice pilot participants complete the Predict, Run R, Explore, and Explain loop without facilitator intervention. This small-sample target will be reported with counts and limitations.
- All critical accessibility defects found during the pilot are fixed or explicitly documented before release.
- At least one contributor outside the core team completes an issue, lesson adaptation, or documentation improvement.

Future work

Future contributors could add domain-specific lesson packs, localization, optional WebXR adapters, additional visualization types, and integrations with established R visualization packages. These extensions remain outside the funded minimum viable product so that the project can deliver a stable and reusable foundation within six months.